



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	GradeMind – AI-Based Academic Performance Prediction & Student Analytics	
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Abstract

Academic institutions routinely generate large volumes of student data—attendance, internal assessment scores, assignment submissions, and attendance patterns—yet this data is rarely used to its full potential for early intervention. GradeMind is an AI-based academic performance prediction and student analytics system designed to address this gap by leveraging machine learning to forecast student outcomes and identify at-risk learners well before final examinations. The system aggregates historical and ongoing academic records, applies feature engineering to capture patterns such as attendance consistency, assignment completion rates, and trends across internal assessments, and trains predictive models—including regression and ensemble-based classifiers—to estimate a student's likely academic performance. A web-based dashboard presents these predictions alongside visual analytics, enabling faculty and academic mentors to identify students who may need additional support and to track cohort-level trends over a semester. By combining predictive analytics with an intuitive interface, GradeMind aims to support data-driven decision-making in academic mentoring, reduce dropout and failure rates through timely intervention, and give students actionable feedback on their own progress. The project also explores the interpretability of the underlying models, ensuring that predictions are accompanied by clear, explainable factors rather than opaque scores, so that both educators and students can trust and act upon the insights generated by the system.

Signature of the Guide

HOD